TYPES OF GALAXY

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THROUGHOUT THE UNIVERSE, galaxies exist in enormous diversity. These vast wheels, globes, and clouds of material vary hugely in size and mass—the smallest contain just a few million stars, and the largest around a million million. Some are just a few thousand light-years across, and others can be a hundred times that size. Some contain only old red and yellow stars, while others are blazing star factories, full of young blue and white stars, gas, and dust. The features of galaxies are clues to their history and evolution, but astronomers have only recently begun to put the entire story together—and there are still many gaps in their knowledge.

EDGE-ON SPIRALS
NGC 1055 is a spiral galaxy that happens to lie edge-on to Earth. This image reveals that its dark dust lanes, silhouetted against the brightness of its stars, have been warped out of shape, probably in a close encounter with another galaxy.

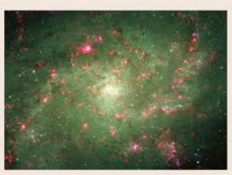
THE VARIETY OF GALAXIES

Galaxies can be classified by their shape, size, and color. At the most basic level, they are divided by shape into spiral, elliptical, and irregular galaxies. Edwin Hubble (see p.45) devised a more precise classification, still used today, that subdivides these galaxy shapes. Hubble classed spiral galaxies as types Sa to Sd—an Sa galaxy has tightly wound spiral arms, while an Sd very loose arms. Spirals with a bar across their center are classed as SBa to SBd. Hubble classed elliptical galaxies as E0 to E7 according to their shape in the sky—circular galaxies are E0, and elongated ellipses E7. Elliptical galaxies appear as two-dimensional ellipses, but in reality, they are three-dimensional ellipsoids ranging from roughly ball-shaped star clouds to cigar shapes. So Hubble's classification does not reflect their true geometry, since an E0 galaxy could be a cigar shape viewed end-on from Earth. Hubble also recognized an intermediate type of galaxy—the lenticular (type S0), with a spiral-like disk, a hub of old yellow stars, but no spiral arms. Finally, irregular galaxies (type Irr) are usually small and rich in gas, dust, and young stars, but have few signs of structure.

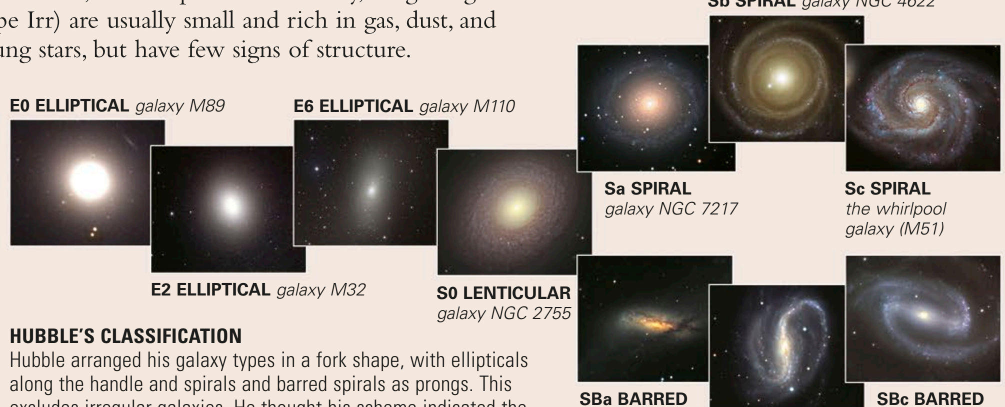
NUMBERING ELLIPTICALS
The class of an elliptical galaxy is found by dividing the difference between its long and short axes by the long-axis length and then multiplying by 10, making this galaxy, M110, an E6.

IRREGULAR GALAXY
Clouds of stars that lack clear disk- or ellipselike structure are called irregular galaxies. The Small Magellanic Cloud is one such irregular galaxy.

ELLIPTICAL GALAXY
Balls of stars, from perfect spheres, through egg shapes (such as M59, pictured here), to cigar-shaped ellipsoids, are called elliptical galaxies.



SPIRAL GALAXY
Vast, rotating disks of stars, dust, and gas are classed as spiral galaxies. Spirals have a ball-shaped nucleus inside a disk with spiral arms. M33 is a nearby spiral galaxy.

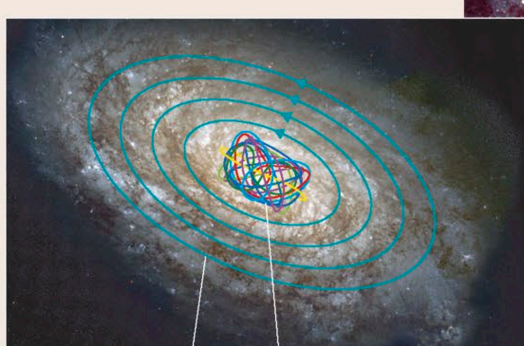


HUBBLE'S CLASSIFICATION
Hubble arranged his galaxy types in a fork shape, with ellipticals along the handle and spirals and barred spirals as prongs. This excludes irregular galaxies. He thought his scheme indicated the evolution of galaxies—today, astronomers know it is not so simple.

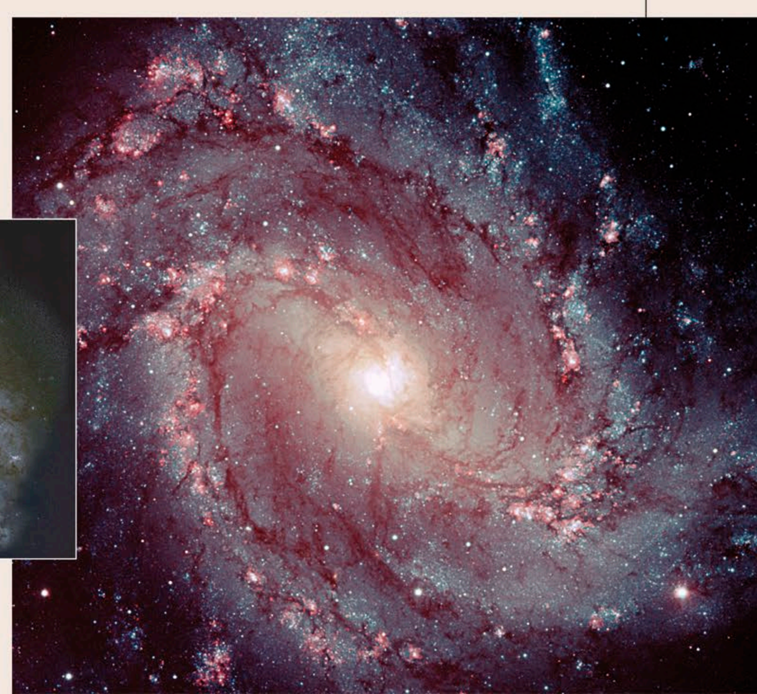
SPIRAL GALAXIES

Some 25–30 percent of galaxies in the nearby universe are spirals. In each one, a flattened disk of gas- and dust-rich material orbits a spherical nucleus, or hub, of old red and yellow stars, which is often distorted into a bar. Stars occur throughout the disk, but the brightest clusters of young blue and white stars are found only in the spiral arms. The space between the arms often looks empty viewed from Earth, but it is also full of stars. Above and below the disk is a spherical "halo" region, where globular clusters (see p.289) and stray stars orbit. Spiral galaxies rotate slowly—typically once every few hundred million years—but they do not behave like a solid object. Stars orbiting farther away take longer to complete an orbit than those close to the core. The resulting "differential rotation" is the key to understanding the spiral arms.

BARRED SPIRAL
Similar to our own galaxy, M83 (right) is a typical barred spiral, having a straight bar on either side of the galactic nucleus.



ORBITS IN SPIRALS
Stars in the disk of a spiral galaxy follow elliptical, nearly circular orbits in a single plane. Those in the hub have wildly irregular orbits at a multitude of angles.



BEYOND THE MILKY WAY